Econophysics: If price changes are random, how do companies and hedge funds make profits?!

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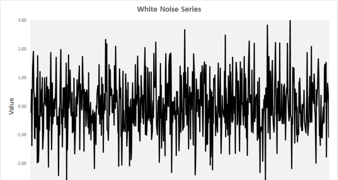
Motivation and Introduction

Motivation

- There are many processes in nature that seem random, such as the weather, the motion of molecules, and the **social behavior of people**.
- The dynamics of financial markets is an **emergent phenomenon** resulting from **complex systems** that involve the interaction of people's decisions.
- In this note, I will discuss(mathmatically) that while price changes appear random, there are certain anomalies within them, and there are algorithms (strategies) to make a profit (free lunch opportunities). However, from a general perspective, market dynamics should be considered random, which is why many traders tend to break even in the long term.



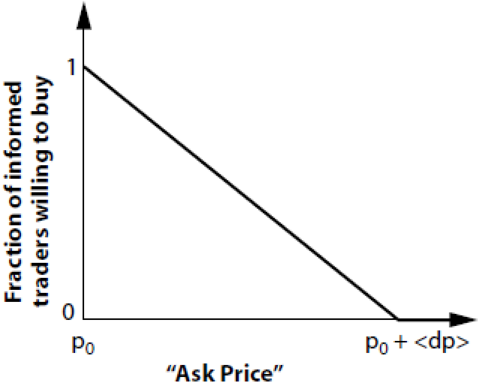
Informed and Noise Traders

Informed and Noise Traders

- **Informed traders** possess access to information that is not yet reflected in current market prices. This information could be related to a company's financial health, upcoming product releases, macroeconomic data, or other material events.
- **Noise traders:**, on the other hand, make decisions based on a mix of random information, speculation, or even misinformation. They may be influenced by trends, market rumors, psychological factors, or misinterpretations of data.

Informed and Noise Traders: a toy model

- Consider P_0 as the last quote, and informed traders regard it as a valuable asset. When the asking price approaches $P_0 + < dp >$, informed traders lose interest in buying.[1]. As you can see, using some simple ideas, we can develop a toy model to understand the market better.



[1] picture is adapted from "Didier Sornette, 2018"

Are Price Dynamics Unpredictable?

Are Price Dynamics Unpredictable?

- Acting on advantageous information adjusts the price, reducing or eliminating the potential gain due to the action's impact on the price. This illustrates the idea that intelligent, informed investor actions contribute to price randomness, as proposed by Bachelier, Samuelson, and others. Conversely, **without informed traders, profit opportunities persist since the buying price remains unchanged at P_0.**

Are Price Dynamics Unpredictable?

- In the absence of noise traders, there would be no counterparts for trades, resulting in no market activity because if everyone agrees on the price, there is no profit to be made. **Therefore, the stock market requires some level of noise to provide liquidity.** Intelligent traders then work hard, and according to this theory, their investments will make the market completely noisy, leaving no intelligible signals.

General Mathematical Proof that Anticipated Prices are Random (Samuelson)

General Mathematical Proof that Anticipated Prices are Random (Samuelson)

- Samuelson[2] has rigorously proved a general theorem showing that the concept of **unpredictable prices can be derived from a model. This model hypothesizes that a stock's current price**, P_t, is the expected discounted value of its future dividends, $d_t, d_{t+1}, \ldots$, which are assumed to be random variables generated by a general (but known) stochastic process:

$$P_t = d_t + \phi_1 d_{t+1} + \phi_1 \phi_2 d_{t+2} + \phi_1 \phi_2 \phi_3 d_{t+3} + \cdots$$

  where $\phi_i = 1 - r < 1$,

[2] Samuelson, P. A. (1965). Proof that properly anticipated prices fluctuate randomly, Industrial Management Review 6, 41–49.

General Mathematical Proof that Anticipated Prices are Random (Samuelson)

- We see the expectation $\mathbb{E}[P_{t+1}]$ of P_{t+1} conditioned on the knowledge of the present price P_t is

$$\mathbb{E}[P_{t+1}] = \frac{P_t - d_t}{\phi_1}$$

- This demonstrates that, apart from the drift caused by inflation and dividends, price increments lack a systematic component or memory of past events, making them random. Consequently, **even when the economy is not entirely free to fluctuate randomly, intelligent speculation can turn observed stock price changes into a random process.**

If Price Changes Are Random, How Do Companies and Hedge Funds Make Profits?

If Price Changes Are Random, How Do Companies and Hedge Funds Make Profits?

- It is a very interesting question, and I will discuss it in another note. But let's have a quick overview. As we have shown, price changes are random, but there are fluctuations or anomalies caused by the interactions of investors or traders and the complex information processing in their decisions. Therefore, it seems likely that there should be rigorous algorithms to find arbitrage opportunities over different time scales, depending on various factors (which I will discuss in my next note). But, **keep in mind that many naive investors or traders, lacking insight and strong algorithms, will, on average, break even in the long term** (also consider some chaotic effect).

Risk-Return Trade-Off

Risk-Return Trade-Off

- One of the central insights of modern financial economics is the necessity of balancing risk and expected return. While Samuelson's version of the efficient markets hypothesis restricts expected returns, it does not consider risk. Specifically, a security's positive expected price change might be the reward needed to attract investors to bear its associated risks. In fact, a sufficiently risk-averse investor might be willing to pay to avoid holding a security with unforecastable returns.

Risk-Return Trade-Off

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